

Special Random Processes

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1 Poisson Process

Definition 1

A random process $\{X(t) \mid t \geq 0\}$ is called a **Poisson process** with rate $\lambda > 0$ if $X(t)$ represents the number of occurrences of an event in $(0, t)$ and the following postulates are satisfied.

Postulates 1

$$P(1 \text{ occurrence in } (t, t+h)) = \lambda h + O(h)$$

Here $O(h)$ contains higher powers of h .

Postulates 2

$$P(2 \text{ or more occurrences in } (t, t+h)) = O(h)$$

Postulates 3

$$P(0 \text{ occurrences in } (t, t+h)) = 1 - \lambda h + O(h)$$

Postulates 4

$X(t)$ is independent of the number of events occurring prior to the interval $(t, t+h)$.

Postulates 5

Probability depends only on interval length and not on starting point:

$$P(\text{events in } (t_1, t_1 + t)) = f(t)$$

2 Birth and Death Process

Let $P_n(t)$ denote probability of n individuals at time t . Let λ and μ represent mean birth and death rates respectively.

Postulates 6 – Birth Rules

$$\begin{aligned} P(1 \text{ birth in } (t, t+h)) &= \lambda h + o(h) \\ P(\geq 2 \text{ births in } (t, t+h)) &= o(h) \\ P(0 \text{ births}) &= 1 - \lambda h + o(h) \end{aligned}$$

Postulates 7 – Death Rules

$$\begin{aligned} P(1 \text{ death in } (t, t+h)) &= \mu h + o(h) \\ P(\geq 2 \text{ deaths}) &= o(h) \\ P(0 \text{ deaths}) &= 1 - \mu h + o(h) \end{aligned}$$

Derivation 1

We determine $P_n(t+h)$ by conditioning on interval $(0, t)$ and $(t, t+h)$:

$$\begin{aligned} P_n(t+h) &= P_n(t)(1-\lambda h)(1-\mu h) + P_{n-1}(t)(\lambda h)(1-\mu h) \\ &\quad + P_{n+1}(t)(1-\lambda h)(\mu h) + P_n(t)(\lambda h)(\mu h) + o(h) \end{aligned}$$

Discarding h^2 terms:

$$P_n(t+h) = P_n(t) - (\lambda + \mu)hP_n(t) + \lambda hP_{n-1}(t) + \mu hP_{n+1}(t) + o(h)$$

Subtract $P_n(t)$ and divide by h :

$$\frac{P_n(t+h) - P_n(t)}{h} = \lambda P_{n-1}(t) - (\lambda + \mu)P_n(t) + \mu P_{n+1}(t) + \frac{o(h)}{h}$$

Take limit $h \rightarrow 0$:

$$\boxed{P'_n(t) = \lambda P_{n-1}(t) - (\lambda + \mu)P_n(t) + \mu P_{n+1}(t), \quad n \geq 1}$$

Result 1

For $n = 0$:

$$P'_0(t) = \mu P_1(t) - (\lambda + \mu)P_0(t)$$

Equations above form the **Kolmogorov forward equation** for a birth-death process.

3 Pure Birth Process

A pure birth process assumes events occur only as “births” and no deaths occur.

Definition 2

A random process is said to be a **Pure Birth Process** if the only possible change in state during $(t, t+h)$ is an increase by one individual and the system never decreases in state.

Postulates 8

For a sufficiently small increment $h > 0$:

$$P(X(h) = 1) = \lambda h + O(h)$$

$$P(X(h) \geq 2) = O(h)$$

$$P(X(h) = 0) = 1 - \lambda h + O(h)$$

and recurrence applies independent of past history.

Derivation 2 — Step 1: Conditioning

Condition on the number of individuals just before t :

$$P_n(t + h) = P[X(t + h) = n].$$

Two mutually exclusive ways yield n individuals at $t + h$:

$$P_n(t + h) = P_n(t) \cdot P(0 \text{ births}) + P_{n-1}(t) \cdot P(1 \text{ birth}) + O(h).$$

Substitute postulate probabilities:

$$P_n(t + h) = P_n(t)(1 - \lambda h + O(h)) + P_{n-1}(t)(\lambda h + O(h)) + O(h)$$

Derivation 2 — Step 2: Simplification

Expand and ignore $O(h^2)$ terms:

$$P_n(t + h) = P_n(t) - \lambda h P_n(t) + \lambda h P_{n-1}(t) + O(h)$$

Subtract $P_n(t)$:

$$P_n(t + h) - P_n(t) = -\lambda h P_n(t) + \lambda h P_{n-1}(t) + O(h)$$

Divide by h :

$$\frac{P_n(t + h) - P_n(t)}{h} = -\lambda P_n(t) + \lambda P_{n-1}(t) + \frac{O(h)}{h}$$

Take the limit $h \rightarrow 0$:

$$\boxed{P'_n(t) = -\lambda P_n(t) + \lambda P_{n-1}(t), \quad n \geq 1}$$

Result 2 — Base Case

For $n = 0$:

$$P'_0(t) = -\lambda P_0(t)$$

with solution:

$$P_0(t) = e^{-\lambda t}.$$

4 Yule–Furry Process

Definition 3

A **Yule–Furry Process** is a special Pure Birth Process where the birth rate is proportional to the current population size:

$$\lambda_n = n\lambda.$$

Postulates 9

Given $N(t) = n$, then:

$$P(\text{one birth in } (t, t+h) \mid X(t) = n) = n\lambda h + O(h).$$

Derivation 3 — Step 1: Forward Equation

Start from pure birth differential system and replace $\lambda \rightarrow \lambda_n = n\lambda$:

$$P'_n(t) = -(n\lambda)P_n(t) + (n-1)\lambda P_{n-1}(t), \quad n \geq 1$$

Base condition: $P_1(0) = 1, P_n(0) = 0 \ (n \neq 1)$.

Derivation 3 — Step 2: Solving $P_1(t)$

$$P'_1(t) = -\lambda P_1(t)$$

Solve:

$$\frac{dP_1}{P_1} = -\lambda dt \Rightarrow P_1(t) = Ce^{-\lambda t}$$

Apply

$$P_1(0) = 1 \Rightarrow C = 1 \Rightarrow P_1(t) = e^{-\lambda t}.$$

Derivation 3 — Step 3: Solving $P_2(t)$

$$P'_2(t) = -2\lambda P_2(t) + \lambda P_1(t)$$

Rewrite:

$$P'_2(t) + 2\lambda P_2(t) = \lambda e^{-\lambda t}$$

Use integrating factor $e^{2\lambda t}$:

$$\frac{d}{dt}(e^{2\lambda t} P_2(t)) = \lambda e^{\lambda t}$$

Integrate:

$$e^{2\lambda t} P_2(t) = \frac{e^{\lambda t}}{\lambda} + C \Rightarrow P_2(t) = e^{-\lambda t}(1 - e^{-\lambda t})$$

(using $P_2(0) = 0 \Rightarrow C = -1$).

Result 3 — General Formula

By induction:

$$P_n(t) = e^{-\lambda t} (1 - e^{-\lambda t})^{n-1}, \quad n \geq 1$$

Mean and variance:

$$E[X(t)] = e^{\lambda t}, \quad \text{Var}[X(t)] = e^{\lambda t}(e^{\lambda t} - 1)$$

5 Renewal Process

Definition 4

A stochastic process $\{N(t), t \geq 0\}$ is called a **Renewal Process** if:

$$N(t) = \max\{n : S_n \leq t\}$$

where

$$S_n = X_1 + X_2 + \cdots + X_n$$

and $\{X_i\}$ are independent and identically distributed (IID) non-negative random variables representing *inter-arrival times*.

Remarks 1

If inter-arrival times follow an exponential distribution, the renewal process becomes a Poisson process.

Definition 5

The **Renewal Function** or **Expected Number of Events** up to time t is:

$$m(t) = E[N(t)]$$

Derivation 4 — Step 1: Conditioning Argument

Condition on the time of the first event X_1 :

$$m(t) = F(t) + \int_0^t m(t-x) dF(x)$$

where $F(t)$ is the CDF of interarrival time.

Result 4 — Fundamental Renewal Equation

$$m(t) = F(t) + \int_0^t m(t-x) dF(x)$$

This is known as the **Renewal Integral Equation**.

Remarks 2

If F has finite mean μ , then as $t \rightarrow \infty$:

$$\frac{m(t)}{t} \rightarrow \frac{1}{\mu}$$

This is called the **Elementary Renewal Theorem**.

6 Summary of Results

Result 5

Poisson Process

$$P[X(t) = n] = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$$

Result 6

Birth–Death Process Forward Equation

$$P'_n(t) = \lambda P_{n-1}(t) - (\lambda + \mu) P_n(t) + \mu P_{n+1}(t)$$

Result 7

Pure Birth Process Equation

$$P'_n(t) = -\lambda P_n(t) + \lambda P_{n-1}(t)$$

Result 8

Yule–Furry Distribution

$$P_n(t) = e^{-\lambda t} (1 - e^{-\lambda t})^{n-1}, \quad n \geq 1$$

$$E[X(t)] = e^{\lambda t}, \quad \text{Var}[X(t)] = e^{\lambda t} (e^{\lambda t} - 1)$$

Result 9**Renewal Equation**

$$m(t) = F(t) + \int_0^t m(t-x) dF(x)$$

Remarks 3

These processes form the foundation for queueing theory, reliability engineering, computational biology, machine maintenance modeling, reservoir analysis, and communication systems.